IS2218 Cheatsheet

AY24/25 - Anselm Long

1 Time Value of Money

1.1 Future Value and Compound Interest

Future Value – amount to which an investment will grow after earning interest.
Simple Interest – interest earned only on the original investment.
Compound Interest – interest earned on interest. Note that compound interest is $FV_{compound} - FV_{simple}$, don't confuse this with FV.

$$FV = C \times (1+r)^t$$

1.2 Present Value

Present Value – value today of a future cash flow.

$$PV = \frac{FV}{(1+r)^t} \quad , \quad PV = DF \times FV$$

Discount Factor (DF) – PV of $1

$$DF = \frac{1}{(1+r)^t}$$

Discounted Cash Flow (DCF) – calculate present value by discounting future cash flows

1.3 Perpetuities and Annuities

Perpetuity – a stream of level cash payments that never ends
To receive cash now, add C to formula. To receive n years from now, divide by $(1+r)^{n-1}$ to get PV.

$$PV = \frac{C}{r}$$

Annuity – level stream of cash flows at regular intervals with a finite maturity
Regular Perpetuity - Delayed Perpetuity

$$PV = C\left[\frac{1}{r} - \frac{1}{r(1+r)^t}\right]$$

Watch out for time in terms of months vs years!
PV Annuity Factor (PVAF) – present value of $1 a year for each of t years

$$PVAF = \left[\frac{1}{r} - \frac{1}{r(1+r)^t}\right]$$

Future Value of Annuity Payment

$$\begin{aligned} FV &= PV \times (1+r)^t \\ &= (C \times PVAF) \times (1+r)^t \\ &= C \times \frac{(1+r)^t - 1}{r} \end{aligned}$$

Annuity Due – cash flows starting immediately (e.g. beginning of the year)

$$PV_{due} = PV_{annuity} \times (1+r)$$
$$FV_{due} = FV_{annuity} \times (1+r)$$

Amortisation Schedule - For every year-end payment, portion is used to pay interest. Principal balance is then reduced by remaining amount, less interest to pay for subsequent payments.
Use Annuity formula with remaining years of loan to find remaining principal balance.

1.4 Effective Interest Rates

Effective Annual Interest Rate (EAR) – interest rate over a year based on compound interest

$$EAR = (1+MR)^{12} - 1$$

Annual Percentage Rate (APR) – interest rate over a year based on simple interest

$$APR = MR \times 12$$

1.5 Inflation

Inflation – rate at which (RAW) prices as a whole are increasing.
Nominal IR (NIR) – RAW money invested grows (current dollar cash flows discounted by NIR)
Real IR (RIR) – RAW purchasing power of an investment increases (real cash flows discounted by RIR)

$$1 + RIR = \frac{1+NIR}{1+IR}, \quad RIR = \frac{1+NIR}{1+IR} - 1$$

$$RIR \approx NIR - IR$$

Convert real to nominal $\rightarrow NIR = RIR \times (1+IR)^t$

2 Platform Business Model

2.1 Advantage of Platforms

No gatekeepers, new sources of value and supply, community feedback loops

2.2 Network Effects

Definition – impact that the number of users has on value created for each user.
Two-sided – e.g. Uber: riders attract drivers and vice versa.
Positive - provides significant value for users, eg. Facebook
Negative – matching difficulty, imbalances
Four kinds:

- Positive Same Side – e.g., telephones
- Negative Same Side – competition
- Positive Cross Side – e.g., Visa, Mastercard
- Negative Cross Side – e.g., OkCupid

2.3 Monetization

- **Transaction Fee** – charge buyers/sellers per interaction, low friction. Platform has to provide value and offer trust (Upwork)
- **Access** – charge producers to reach users, e.g., LinkedIn
- **Enhanced Access** – pay to boost presence, e.g., Google Ads, dating apps
- **Enhanced Curation** – higher-quality interactions, e.g., vetting/screening
- **Monetization Principles**
  - Don't charge for value users got free before
  - Don't restrict access to accustomed value
  - Create new value that justifies pricing
  - Consider monetization in design stage

2.4 Metrics

Startup Phase

- Liquidity – listings leading to interactions
- Matching Quality – conversion rate accuracy
- Trust – risk comfort level

Growth Phase

- Producer – frequency, outcomes, retention
- Consumer – usage, searches, conversion rate

Maturity Phase – innovate, absorb standalone apps (e.g., Microsoft)

2.5 Future of Platform Revolution

Disruptable Industries - Info-intensive, fragmented, or with info asymmetry
Resistant Industries - Heavily regulated, high failure cost, resource intensive

3 Coursera - Platform Thinking

Platforms' core mechanism - Idle Assets Hunters (eg. Uber)

3.1 Types of Platforms

- **Product Platforms** - A set of components that create a basic structure common to many products (Sony Walkman)
- **Innovation Platforms** - Act as a foundation on which external innovators can develop their own complementary products (PC by Microsoft, Intel, IBM, Operating Systems)
- **Transactional Platforms** - products or services where two or multiple groups of customers get together through a platform that internalizes indirect network externalities (Visa, MasterCard, App Store)
- **Orthogonal Platforms: Client-as-a-Target** - products or services based on sale of different services to two groups of customers who do not come into direct contact with each other, rely on unidirectional cross side network externality (Newspaper Advertising)
- **Orthogonal Platforms: Client-as-a-Source** - same as above, but client data is collected (Twitter)
- **Hybrid Platforms** - different definitions of platforms can co-exist, with transactional and orthogonal relationships, with client as a target & source (Facebook & Apple)

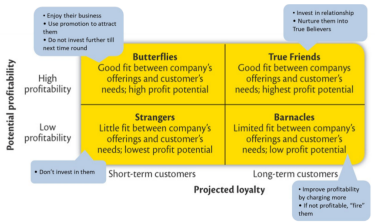
4 Marketing

Marketing is the activity, set of institutions and processes for creating, communicating, delivering, and exchanging offerings that have value for customers, clients, partners, and society at large.

4.1 Selling Concepts

- **Production Concept** - Focus on production and distribution efficiency, assuming consumers prefer products that are available and highly affordable
- **Product Concept** - Make continued product innovations, assuming consumers want products with highest quality, best features, and performance
- **Selling Concept** - Concentrate on selling, assuming consumers will not buy enough unless stimulated aggressively
- **Marketing Concept** - determine needs and wants of target markets and deliver them more effectively than competitors. Create, build & maintain beneficial exchanges with target buyers. Recognize needs.

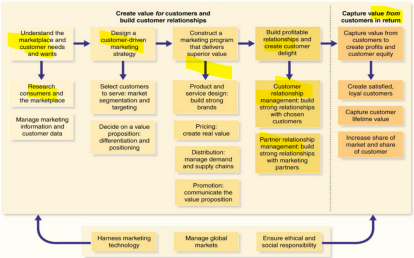
Customer Relationship Management - process of building and maintaining profitable customer relationships by delivering superior customer value and satisfaction



4.2 Marketing Management Process

1. **Analyze & Identify Market Opportunities**
2. **Research & Select Target Markets** - Bottom of the Pyramid (earliest stages of economic development)
   - **Segmentation** - dividing a market into smaller groups with distinct needs, characteristics, or behaviors who might require separate products or marketing
   - **Targeting** - process of evaluating each market segment's attractiveness and selecting one or more segments to enter
   - **Positioning** - arranging for product to occupy distinctive and desirable place relative to competing products in consumers mind
3. **Develop Marketing Mix (4Ps)** - Product, Price, Promotion, Place
4. **Managing the Marketing Effort** - Analysis, planning, implementation, control

Takeaways
Understand your customer (needs and trends), know your company (strengths and weaknesses), analyze your competition (threats).



5 Customer Lifetime Value

5.1 Delivering Customer Value

Be the best at one, maintain industry standards in the other two.

- **Operational Excellence** - reliable products at competitive prices and delivered with minimal difficulty
- **Product Leadership** - offering leading-edge products that renders rivals' goods obsolete
- **Customer Intimacy** - segmenting and targeting markets precisely and tailoring offerings to match exactly the demands of those niches

5.2 CLV

Definition - Dollar value of a customer relationship, net present value of all future cash flows from a customer relationship

$$CLV = (M - R)(\frac{1+d}{1+d-r})$$

Where M - margin = revenue - variable costs, R - retention spending, r - retention rate, d - discount rate.
For businesses where revenue is obtained after service is delivered:

$$CLV = (M - R)(\frac{r}{1+d-r})$$

Observations

- Low retention rate & high discount rate → most of CLV achieved in first few periods
- High retention rate & low discount rate → little CLV was achieved in the first few periods.
- Retention rates increase over time - calculate CLV among customers of same cohort.
- Fire customers with negative CLV.

Customer Acquisition - spend less than CLV
Customer Retention - spend less than difference in CLV
Survival Rate - Probability that customer has a relation in a given period: $t = r^{t-1}$

$$CLV_{Segment} = CLV_{AvgCustomer} \times N$$

N - number of customers in segment

6 Pricing

Definition - sum of values consumers exchange for benefits of having or using product

6.1 Internal Factors determining Price

- **Marketing Objectives** - survival, profit max, market share, quality, cost recover, etc.
- **Marketing-Mix Strategy** - product design, distribution, promotion, curbing parallel imports
- **Costs** - costs are the floor price, demand is the ceiling price

6.2 External Factors determining Price

- **Consumer Perceptions** - Price-Quality Relationship
- **Competitors** - reference point, product homogeneity
- **Other** - economic conditions, government

6.3 Experience and Experience Curves

Experience - combined effects of learning, volume, investment, and specialization
Experience Curve - illustrates relationship between cumulative production & per unit costs. Drives prices down

6.4 Methods and Strategies

- **Product Pricing Strategies**
  - **Price Skimming** – set a high price, company makes fewer but more profitable sales. Use when segmentation on price elasticity, no immediate competition
  - **Price Penetration** – set a low initial price, attract large number of buyers quickly and win large market share. Use when consumers are price sensitive, have experience curve effects, and with competition
- **Cost-Oriented Strategies**
  - **Markup Pricing based on Cost** – predetermined percentage markup
  - **Target Profit Pricing** – predetermined return on capital used to produce and market profit
  - Let P be the price, DVC be Direct Variable Cost, FC is Fixed Cost, X is the standard volume, r is Desired Rate of Return, and K is capital used.

    $$P = DVC + \frac{FC}{X} + \frac{rK}{X}$$

    This approach requires good demand estimates.
- **Demand Oriented Pricing** – Based on consumer perceptions and behavior
- **Perceived-Value Pricing** – Price based on buyer's perception of value
- **Odd or Psychological Pricing** – Set prices below even-dollar amounts
- **Loss-Leader Pricing** – Set prices of selected products at lower levels to increase sales of others
- **Optional-Product Pricing** – Sell accessory products with main product
- **Captive-Product Pricing** – Products that must be used with main product are expensive while the main product is cheap
- **Bundled Pricing** – Products combined and sold as bundle at reduced price
- **Price Discrimination** – Peak pricing, student pricing etc.